Universitatea Transilvania din Brașov

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FACULTATEA DE INGINERIE ELECTRICĂ ȘI ȘTIINȚA CALCULATOARELOR

Department of Electronics and Computing
Study Programme: Applied Electronics

Verification Project
Smart Lamp Controller

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1 THE DIGITAL DESIGN OF THE CIRCUIT

1.1 BLOCK DIAGRAM

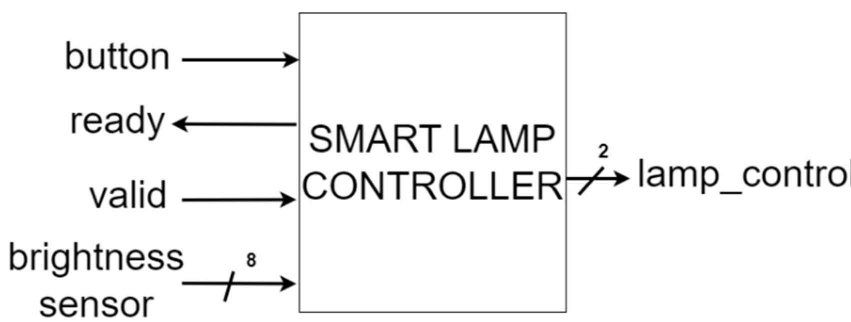

Figure 1. Block diagram of the Smart Lamp Controller

1.2 DESCRIPTION OF THE GENERAL FUNCTIONALITY OF THE DUT (DEVICE UNDER TEST)

The DUT controls the light intensity of a lamp. Its functionality depends on two modes: manual mode and automatic mode. The initial state of the lamp is off. A short press from this state switches directly to automatic mode, and a long press when the lamp is off, switches to manual mode. The lamp switches between the two modes by long pressing its button for minimum 20 clock cycles.

In manual mode, the light intensity changes between 4 states by short pressing the button for less than 20 clock cycles, starting with the lamp switched off. In automatic mode, the DUT changes the light intensity based on the values received from a light sensor. The DUT has 4 pre-defined ranges, representing the lamp illumination steps. The sensor data is transmitted using the valid-ready protocol. The DUT only receives the value from the sensor when the valid signal is active. When the valid is not active, the DUT keeps the previously read value.

1.3 DUT INTERFACES

Table 1. System signals generated in the testbench

General Signals			
Signal Name	I/O	Size	Description
clk	I	1	Clock signal
reset	I	1	Asynchronous reset, active low

Table 2. Sensor Interface

Sensor Interface			
Signal Name	I/O	Size	Description
brightness	I	8	Environmental brightness
valid	I	1	Valid signal
ready	O	1	Ready signal

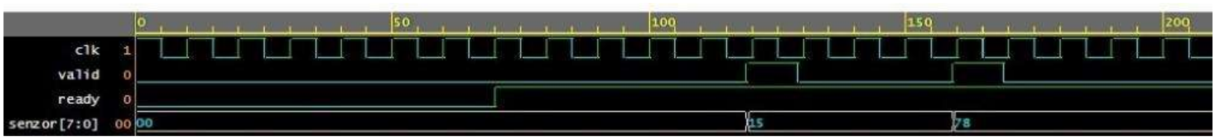

Figure 2. Valid-ready protocol

Table 3. Button Interface

Button Interface			
Signal Name	I/O	Size	Description
button	I	1	For changing modes and states, active low

Table 4. Lamp Interface

Lamp Interface			
Signal Name	I/O	Size	Description
lamp_control	O	2	Lighting thresholds

1.4 List of DUT Functionalities

- Reads button state
- Counts the number of clock cycles when the button is pressed
- Receives environmental brightness from light sensor in automatic mode
- Calculates lamp's light intensity based on press length and sensor data

Table 5. Brightness ranges

Light Intensity	Sensor Values Range	DUT Output Value
On 2	0-63	11
On 1	64-127	10
On 0	128-191	01
Off	192-255	00

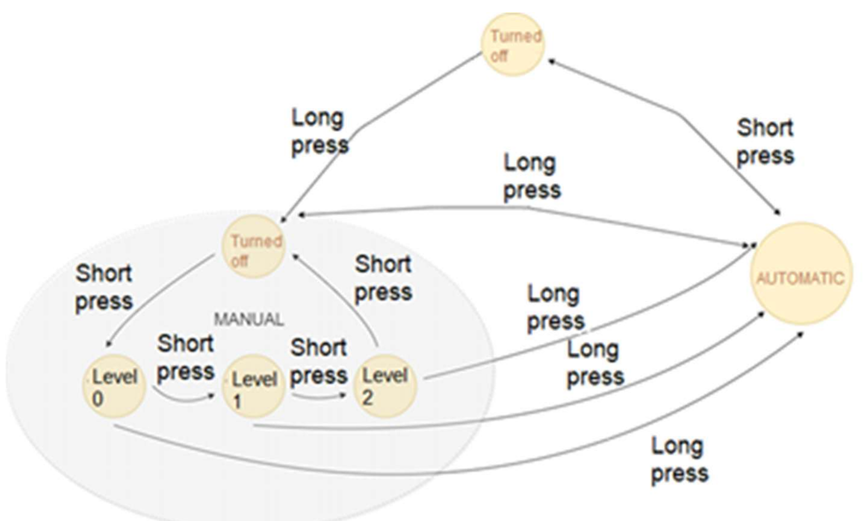

Figure 3. Finite State Machine describing the functionality of the smart lamp

1.5 SCHEMATIC DIAGRAM

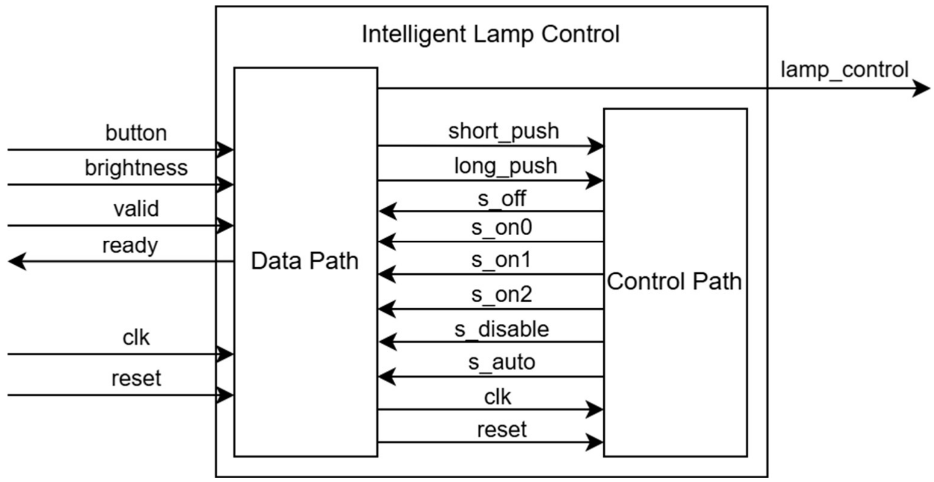

Figure 4. DUT Block Diagram

Table 6. Internal signals of DUT

Signal name	Description
short_push	Indicates that the button has been pressed short
long_push	Indicates that the has been pressed long
s_off	Indicates that the DUT is in the OFF state, lamp is off.
s_on0	Indicates that the DUT is in ON0 state.
s_on1	Indicates that the DUT is in ON1 state.
s_on2	Indicates that the DUT is in ON2 state.
s_disable	Indicates that the DUT is turned off.
s_auto	Indicates that the DUT is in the Automatic mode.

1.6 WAVEFORMS

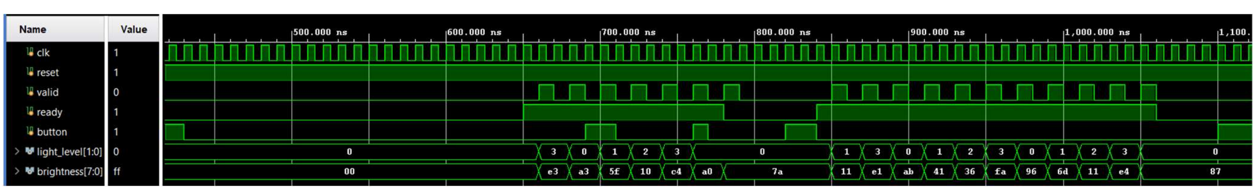

Figure 5. Waveforms from the DUT in automatic mode

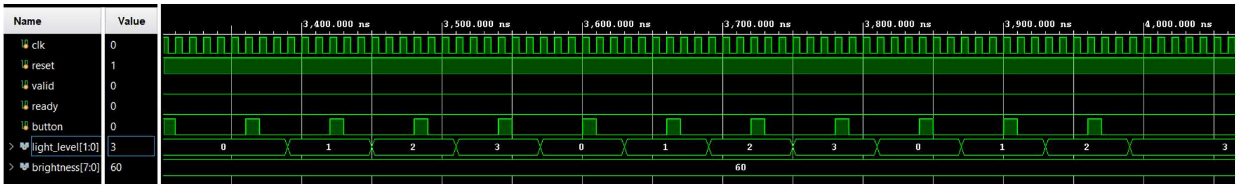

Figure 6. Waveforms from the DUT in manual mode

2 Digital Circuit Verification

2.1 Verification Environment Architecture

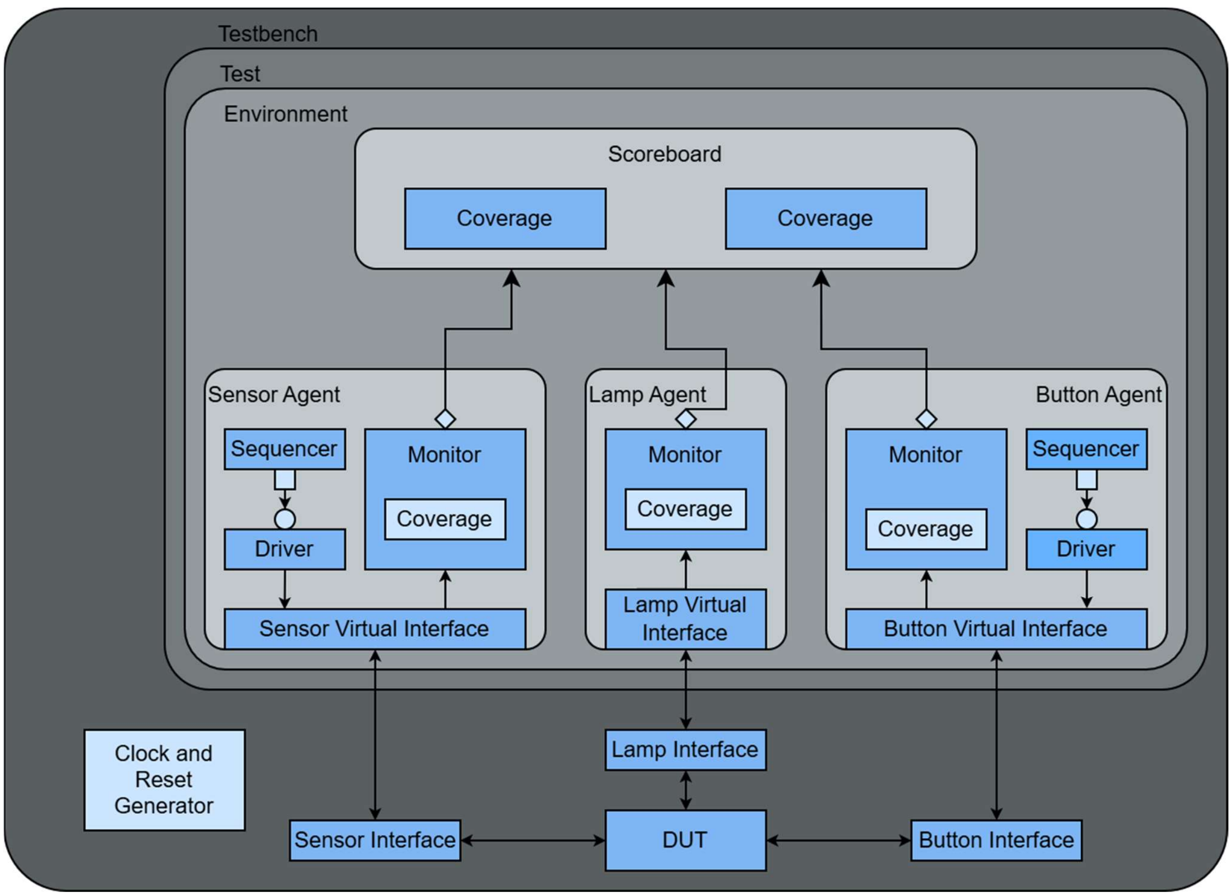

Figure 7. Overview of the components necessary for sending stimuli to the DUT

2.1.1 Sensor Agent

The Sensor Agent is an active agent that encapsulates the Sequencer, Driver and Monitor into a single entity, connecting the components with the DUT using the virtual interface.

2.1.2 Sensor Transaction

Table 7. Sensor Transaction

Field Name	Type	Size	Values	Description
sensor	rand bit	8	[0:255]	Value of environmental brightness

2.1.3 Sensor Driver

The Sensor Driver interacts with the DUT via the Sensor Interface, sending the transactions generated by the Sensor Sequencer. The transactions follow the valid-ready protocol.

The driver receives transactions from the sequencer, waits for the ready signal to assert, asserts valid signal and drives brightness signals, and after a clock cycle de-asserts the valid signal.

2.1.4 Sensor Monitor

The Sensor Monitor waits for valid and ready signals to be both asserted, samples the brightness signal from the Sensor Virtual Interface, reconstructs the transaction and sends it to the Scoreboard and Sensor Coverage Collector.

2.1.5 Sensor Coverage Collector

The Sensor Coverage Collector is instantiated inside the Sensor Monitor and receives transactions from it. The coverage collector contains sensor_cg cover group with sensor_state cover point. It has bins for every range of environmental brightness and for lower limits and upper limits of every range.

2.1.6 Sensor Sequences

The Sensor Sequences generate stimulus for the Sensor Driver. Each sequence creates one or more constrained-random transactions that the driver sends to Sensor Interface. The following sequences are implemented:

Table 8. Sensor Sequences

Sequence	Description
balanced_data_sequence	Sends values of every environmental brightness range equally.
sensor_limit_values_sequence	Sends only lower and upper limits of every environmental brightness range.
sensor_sequence	Sends random values of environmental brightness.

2.1.7 Lamp Agent

The Lamp Agent is a passive agent that contains only the Monitor and connects it with the DUT using the virtual interface. The Lamp Interface has only output signals from DUT, meaning no driver or sequencer is needed.

2.1.8 Lamp Transaction

Table 9. Lamp Transaction

Field Name	Type	Size	Values	Description
light_level_out	rand lamp_state	2	{LIGHT_OFF, ON_LEVEL_0, ON_LEVEL_1, ON_LEVEL_2}	Lamp's light intensity

2.1.9 Lamp Monitor

The Lamp Monitor samples the light level signal every time its value changes, reconstructs the transaction and sends it to the Scoreboard and Lamp Coverage Collector.

2.1.10 Lamp Coverage Collector

The Lamp Coverage Collector is instantiated inside the Lamp Monitor and receives transactions from it. The coverage collector contains lamp_state_cg cover group with lamp_state cover point. It has bins for every light level of the lamp.

2.1.11 Button Agent

The Sensor Agent is an active agent that encapsulates the Sequencer, Driver and Monitor into a single entity, connecting the components with the DUT using the virtual interface.

2.1.12 Button Transaction

Table 10. Button Transaction

Field Name	Type	Size	Values	Description
button	rand button_state	2	{UNPUSHED_BUTTON, SHORT_PUSH_BUTTON, LONG_PUSH_BUTTON}	Type of button press

2.1.13 Button Driver

The Button Driver receives transactions from the Button Sequencer, de-asserts the button signal, waits for several clock cycles based on the press type and asserts the button signal.

2.1.14 Button Monitor

The Button Monitor waits for the button signal to de-assert, counts the number of clock cycles during which it stays low, decides the type of button press, reconstructs the transaction and sends it to the Scoreboard and Button Coverage Collector.

2.1.15 Button Coverage Collector

The Button Coverage Collector is instantiated inside the Button Monitor and receives transactions from it. The coverage collector contains button_cg cover group with button_state cover point. It has bins for every type of button press.

2.1.16 Button Sequences

The Button Sequences generate stimulus for the Button Driver. Each sequence creates one or more constrained-random transactions that the driver sends to Button Interface. The following sequences are implemented:

Table 11. Button Sequences

Sequence	Description
auto_mode_sequence	Button is only shortly pressed to enable and disable automatic mode.
manual_mode _sequence	First push button is long to enable manual mode. The rest are short to switch between lamp light intensities.
mode_switch _sequence	Button is only long pressed to switch between automatic mode and manual mode.
one_long_push_button _sequence	Button is pressed long one time. Used after manual_mode_sequence in case Sensor Driver has remaining transactions.
random_sequence	Random type of button press.

2.1.17 Scoreboard

The Scoreboard receives transactions from all monitors.

When receiving a Button Transaction, the scoreboard predicts DUT's operating mode and the next lamp light level if DUT is in manual mode.

When receiving a Sensor Transaction, the scoreboard uses it to predict the next Lamp Transaction when DUT is in automatic mode.

All predicted Lamp Transactions are stored in a queue.

When receiving a Lamp Transaction, the scoreboard compares it with the predicted transaction, generating an error message if the checker fails.

2.1.18 Scoreboard Coverage Collector

The Scoreboard Coverage Collector is instantiated inside the Scoreboard and receives transactions from it. The coverage collector contains lamp_transition_coverage_cg cover group with 4 cover points: next_lamp_state, current_lamp_state, current_mode, next_mode; and 3 crosses: lamp_transition, operation_transition, lamp_operation_transition.

2.2 VERIFICATION CHECKLIST

2.2.1 Assertions

All Sensor assertions are disabled when reset is active to prevent false assertion failures.

Table 12. Sensor Interface Assertions

Assertion	Description
valid_ready	Verifies that valid signal is asserted only when ready signal is asserted.
valid_puls	Verifies that valid signal is a single pulse.

2.3 FUNCTIONAL COVERAGE ELEMENTS

Table 13. Sensor cover group

sensor_cg			
Cover point	Bins	Values	Description
sensor_state	off	[193:254]	Lamp light off
	on0	[127:190]	Lamp light in ON0 state
	on1	[65:126]	Lamp light in ON1 state
	on2	[1:62]	Lamp light in ON2 state
	limit_value_0	0	Lower limit for light in ON2 state
	limit_value_1	63	Upper limit for light in ON2 state
	limit_value_2	64	Lower limit for light in ON1 state
	limit_value_3	127	Upper limit for light in ON1 state

	limit_value_4	128	Lower limit for light in ON0 state
	limit_value_5	191	Upper limit for light in ON0 state
	limit_value_6	192	Lower limit for light off
	limit_value_7	255	Upper limit for light off

Table 14. Lamp cover group

lamp_state_cg			
Cover point	Bins	Values	Description
lamp_state	LIGHT_OFF	LIGHT_OFF	Lamp light off
	ON_LEVEL_0	ON_LEVEL_0	Lamp light in ON0 state
	ON_LEVEL_1	ON_LEVEL_1	Lamp light in ON1 state
	ON_LEVEL_2	ON_LEVEL_2	Lamp light in ON2 state

Table 15. Button cover group

button_cg			
Cover point	Bins	Values	Description
button_state	UNPUSHED_BUTTON	UNPUSHED_BUTTON	Button unpushed
	SHORT_PUSH_BUTTON	SHORT_PUSH_BUTTON	Button short pressed
	LONG_PUSH_BUTTON	LONG_PUSH_BUTTON	Button long pressed

Table 16. Scoreboard cover group

lamp_transition_coverage_cg	
Cover point	Description
next_lamp_state	Covers the next predicted light intensity
current_lamp_state	Covers the current light intensity
next_mode	Covers the next predicted operation mode

current_mode	Covers the current operation mode
lamp_transition	Covers the transition between light intensities
operation_transition	Covers the transition between operation modes
lamp_operation_transition	Covers the transition between light intensities in both operation modes

2.4 VERIFICATION SCENARIOS AND TESTS

Table 17. Verification Scenarios

Scenario	Description
Sensor Interval Verification	Random values corresponding to each interval are applied while the DUT operates in automatic mode.
Boundary Value Verification	Assign fixed boundary sensor values defined in the specification table.
Continuous Button Press Verification	Button is pressed continuously for long durations to switch between operation modes.
Short and Long Button Press Verification	Use representative clock cycle values for short and long presses.

Table 18. List of implemented tests for DUT verification

Test file name	Test description
auto_mode_test	Test that DUT switches correctly to automatic mode.
manual_mode_test	Test that DUT switched between lamp light levels correctly in manual mode.
long_push_button _test	Test that DUT switches between operational modes correctly.
sensor_limit_values _test	Test that DUT sets the lamp light level correctly when sensor sends boundary values.

sensor_test	Test that DUT sets the lamp light level correctly based on the environment brightness received from the sensor in automatic mode.
global_test	Test that DUT behaves correctly under randomly generated scenarios.

2.5 Results

Running the regression produced the following results:

- Sensor functional coverage: 100%
- Lamp functional coverage: 100%
- Button functional coverage: 100%
- Scoreboard coverage: 100%

Sensor assertions passed without failures throughout the simulation